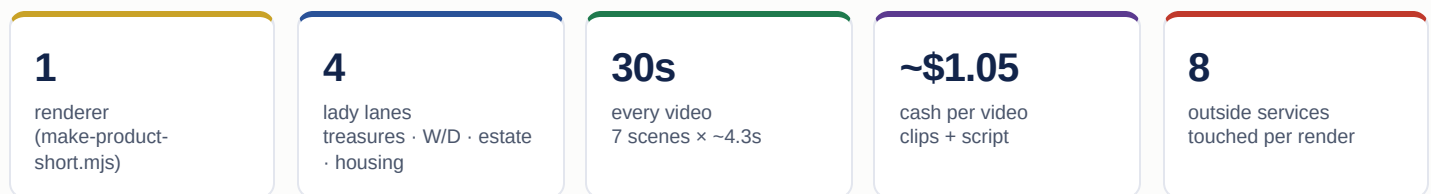


The Lady Video System

One renderer, four businesses, ~1,100 lines of decisions — drawn out so you can change it · August 18, 2026

Every "lady video" you have ever posted — the Treasure Haul product ads, the washer/dryer spots, the estate-sale spots and the daily KC housing brief — is produced by **one program**: `growth-engine/shorts/make-product-short.mjs`. The four businesses are not four pipelines. They are four **briefs** handed to the same renderer, each with its own clock, its own Facebook page and its own link. This document shows that one program end to end: what it touches, in what order, what it costs, what leaves your building, and every knob you can turn.



What this document answers

- **What happens, in order** — the nine stages, with the real artifact each one produced on a real run.
- **What the viewer sees, second by second** — a 30-second video pulled apart at 8 timestamps, with the system that made each frame.
- **Every call that leaves the DGX** — host, purpose, whether it costs money, and where its key lives.
- **How every link in every caption is built** — the affiliate tag, the shelf, the fit code, the UTM.
- **Where the money goes** and **which knob changes what**.

The three sentences version

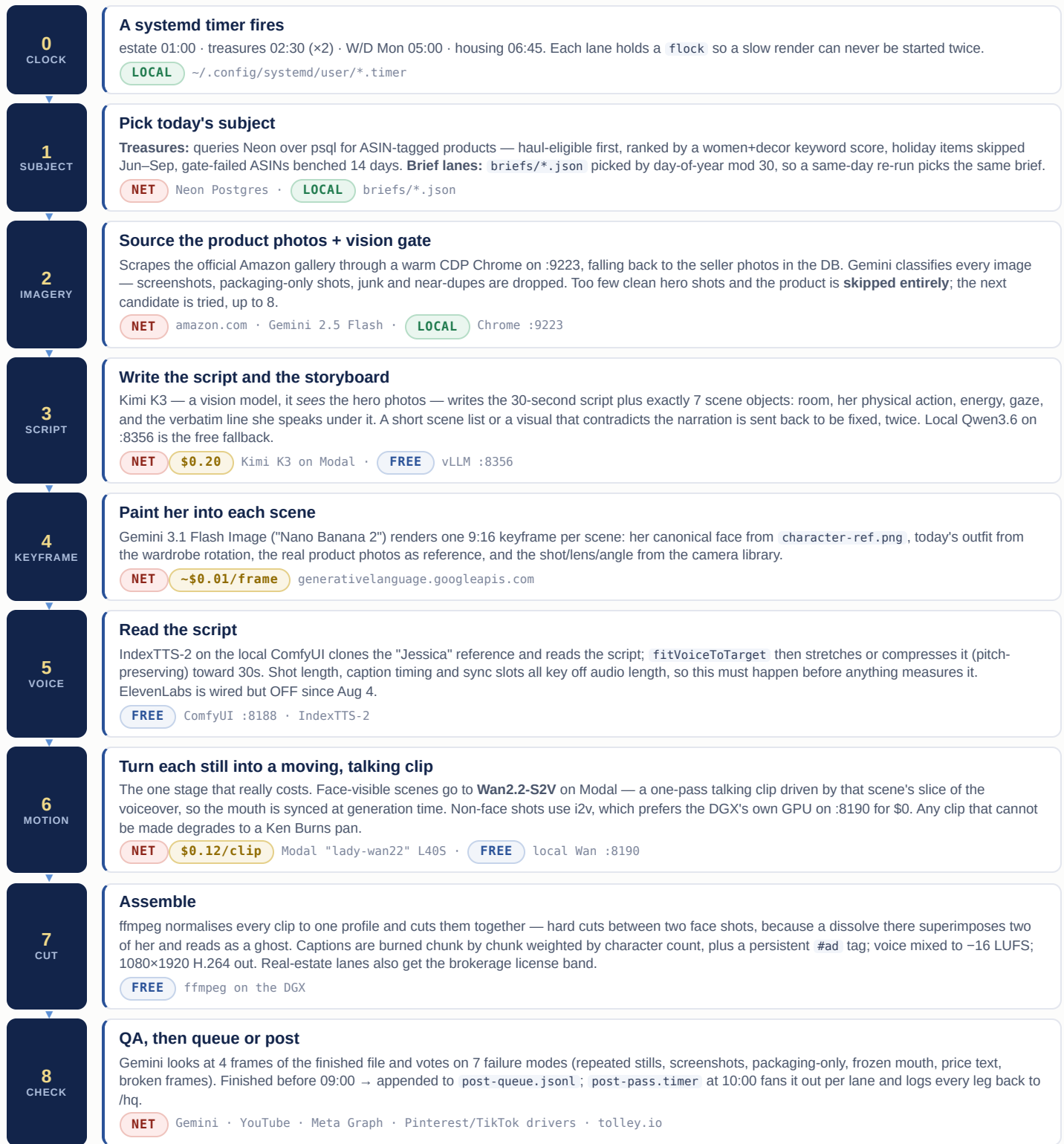
- 1. Night.** A systemd timer fires between 01:00 and 06:45. It picks today's subject — a product out of the shop database, or one of 30 rotating briefs — and hands it to the renderer.
- 2. Render.** Kimi writes a script and a 7-scene storyboard, Gemini paints a keyframe of her in each scene, a local voice clone reads the script, Wan animates each keyframe into a talking clip, ffmpeg cuts it together with burned captions. 1–2½ hours.
- 3. Morning.** Because it finished before 09:00, the video is *queued*, not posted. At 10:00 `post-pass.timer` fans it out to the right pages for that lane and logs the result back to /hq.

The one thing to understand before anything else

Briefs vs. products. The renderer has two front doors. `--auto` means "go find a product in the Neon shop database and sell it" — that is the Treasure Haul lane, and it is the only lane that carries Amazon affiliate links. `--brief file.json` means "here is a business, a pitch, an offer and a link — sell that" — that is W/D, estate and housing, and those captions carry *your own* tolley.io links, no `#ad`, no affiliate tag. Same face, same voice, same 30 seconds, completely different money model. Everything downstream of the script branches on that one flag.

01 The whole machine, top to bottom

Nine stages. **LOCAL** / **FREE** stays on the DGX · **NET** leaves the building · **PAID** costs cash.



Read the arrows as "degrades to", not "stops". No Kimi → Qwen writes it. No keyframe → a product hero fills the slot. No Wan clip → Ken Burns. QA fail → it still renders, it just refuses to auto-post. That is why the lane almost never goes dark — and why a quietly degraded video can ship looking fine in the log.

02 One video, second by second

Real file: shorts/review/2026-08-18-pull-out-cabinet-organizer-pan-sweep.mp4 — rendered 03:52 this morning, 30.00s, 1080×1920, 30fps, template pan-sweep , voice Jessica (IndexTTS-2), 7 scenes, QA clean, \$0.84 of Modal clips.

0:00–0:04
HOOK · scene 1

0:04–0:08
scene 2

0:08–0:12
scene 3

0:12–0:17
scene 4

0:17–0:21
scene 5

0:21–0:25
scene 6

0:25–0:30
CLOSE · scene 7

0:00.5
Scene 1 = the hook. She is already mid-motion — the storyboard forbids a scene that starts from rest.

0:04.5
Scene 2. Hard cut, not a dissolve: two adjacent face shots would ghost across each other.

0:09.0
Scene 3. Same kitchen — the writer is told to change rooms at most twice in a whole video.

0:13.5
Scene 4. Her hands are on the product; the line she speaks here names what she is doing.

0:18.0
Scene 5. The honest-caveat beat. Wan2.2-S2V drives her mouth from this slice of the voiceover.

0:22.5
Scene 6. Captions are re-timed per chunk by character count, not fixed intervals.

0:27.0
Scene 7 = the close. Every video starts and ends on her — no product-still bookends.

0:29.5
The fit CTA tail: "Like the fit? The link for that is below too."

What is actually on the screen

Layer	Made by	When
Her face & body	Gemini 3.1 Flash Image, seeded from character-ref.png	stage 4
Her outfit	Wardrobe rotation — look #599 of persona-wardrobe.json , 1-in-4 chance of a glam look	stage 4
The product	The real seller/Amazon photos, passed to Gemini as reference images	stage 2 → 4
The room	Written by Kimi — told to put the product where it actually lives (a kitchen gadget on a counter, not a default living room)	stage 3
The movement	Wan2.2-S2V on a Modal L40S, one clip per scene, ~69 frames at 16fps	stage 6
The mouth	S2V again — the voiceover slice is the driver, so no separate lip-sync pass runs	stage 6
The voice	IndexTTS-2 cloning the Jessica reference, locally, \$0	stage 5
The yellow captions	ffmpeg drawtext , one text file per chunk, timed by character weight	stage 7
The #ad corner	ffmpeg, burned for the whole duration (FTC + Amazon Associates)	stage 7

Why 7 scenes and not 5 or 12

It is **derived, not chosen**. Wan caps a clip at 81 frames at 16fps ≈ **5.06 seconds**. If a scene slot is longer than one Wan clip, the tail gets clone-held and you see a visible freeze at the end of every scene. So the renderer computes $\text{ceil}(31 \div 5.06) = 7$ scenes from the 30-second target and refuses to let you pin it.

Change the length, and the scene count moves with it — 45s → 10 scenes, 88s → 18. That is also why cutting 45s → 30s on Aug 8 cut roughly a third of the cash cost: fewer paid clips, fewer paid keyframes.

The spoken script, verbatim

"I expected a headache. I got joy. I thought assembly would be a nightmare. No drilling needed. That alone won me over. I slid it into my cabinet. Adjustable width fits perfectly. One drawer glides out smooth. No more digging for buried items. It does take up some depth. **But the access is worth it.** Total game changer for my kitchen. Like the fit? The link for that is below too."

76 words. The arc is enforced by the prompt: hook → the doubt → unboxing → a real usage moment → the honest caveat → the verdict. The caveat is deliberate — it is what makes the rest read as a real person.

03 The same four seconds, at every stage

Pulled out of a real working directory left on disk from the Aug 10 run of the \$100 folding-table product (/tmp/short -EdIyZh). This is literally what the pipeline hands to the next stage.



STAGE 2 • INPUT

The seller photo out of the shop database. Real product, real listing image — nothing generated yet.



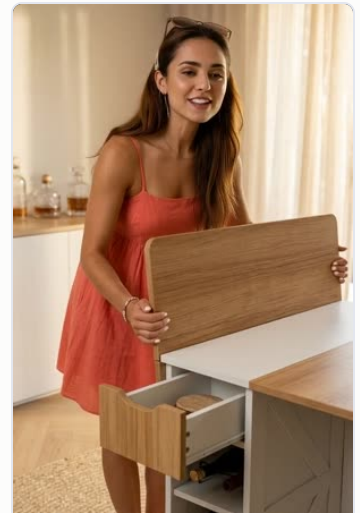
STAGE 4 • KEYFRAME

Gemini renders her into scene 1 holding that product, with her canonical face and today's outfit. A still.



STAGE 6 • CLIP, f.1

Wan takes the keyframe as frame 1 and generates ~69 frames of motion from the scene's "motion" prompt.



STAGE 6 • CLIP, last

The last frame of the same clip. She has moved, the room held, the product survived — that is the whole test.

What the vision gate throws away

Before any of the above, every candidate image is sent to Gemini and labelled. The classes that get dropped: **screenshots** (a phone screenshot of a listing — the July 11 incident this gate was built for), **packaging-only** shots of a closed box, **junk**, and **near-duplicates** of an image already kept.

A product that cannot survive the gate is **benched for 14 days** in `gate-failed.json` and the renderer moves on. Before that bench existed the picker retried the same 8 losers nightly while 70+ untried products waited, and the lane rendered 0 of 2 twice.

The two image sources, in order

- **Official Amazon gallery** for the ASIN, scraped through a persistent Chrome you keep warm on :9223 . The gate is run in `--no-aspect` mode here — gallery images are never phone screenshots, and the aspect rule was killing valid tall hero shots.
- **Seller photos** from the `Product.imageUrl` column in Neon, downloaded and fprobe-validated for decodability before use.

The face is a file, not a prompt



character-ref.png

The canonical portrait. Locked Aug 10.



scene 2 keyframe

Same face, new room, new angle.



amz0.jpg

Official gallery shot, post-gate.

This is the consistency mechanism. Gemini is handed `character-ref.png` on every single scene of every single video across all four lanes, and told: take the face, hair and skin from this photo, take the *clothes* from the text prompt. That split is what lets the wardrobe rotate daily while she stays recognisably the same person.

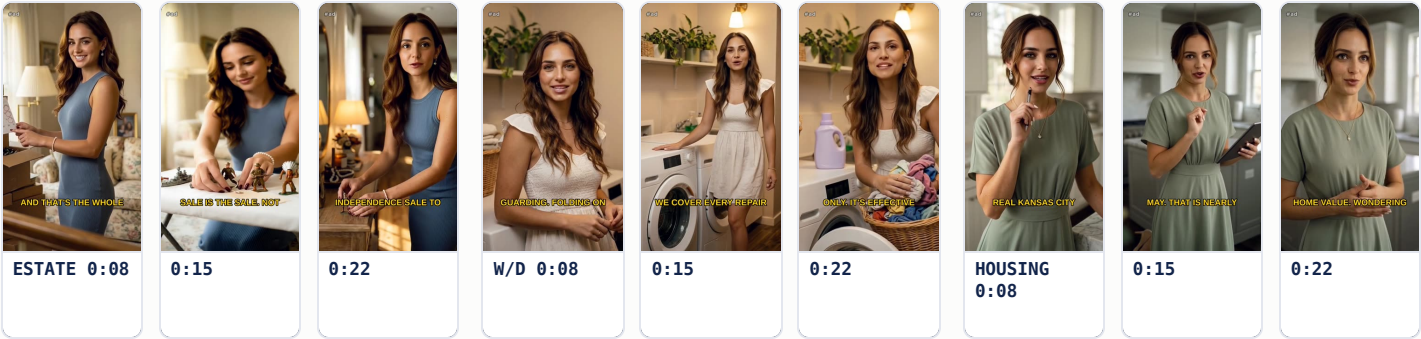
Change that one PNG and every lane changes at once. There is no per-lane face.

The camera is separate again: framing + lens + angle are chosen per scene from `persona-cinema.json` and go into the *keyframe*; the camera *move* goes into the Wan prompt. Kimi is explicitly forbidden from writing camera direction, because its copy fights the choreographed move.

04 The four lanes

Same renderer, same face, same voice. What differs: the clock, the subject source, the link in the caption, and where it gets posted.

Lane	Fires	Subject comes from	Link in the caption	Posts to
 Treasures "product shorts"	02:30 daily ×2 videos	Neon Product table, ASIN-tagged, haul-eligible first	Amazon affiliate /dp/<ASIN> + storefront shelf + tolley.io/shop + fit link	YouTube (@ruthann) · FB Treasure Haul · IG · Pinterest (:9108) · TikTok (:9106) · Bluesky · Threads · X
 Estate	01:00 daily	estate/briefs/*.json — 30 briefs, day-of-year rotation	tolley.io/estate + fit link. No affiliate, no #ad .	Facebook Reel → Tolley Estate Sales page 1192241847311343
 Washer/Dryer	Mon 05:00 weekly	wd- content/briefs/*.json — 30 briefs, day-of-year rotation	tolley.io/wd + fit link	YouTube (@ruthann) + FB Wash & Dry Rental KC 1060351927154451
 Housing "KC Housing Daily"	06:45 daily	Built fresh each morning from the market-pulse brief — not a static file	tolley.io/housing + fit link	YouTube (@yourkchomes) · FB Your KC Homes 230414410149647 · LinkedIn · Pinterest



 **Aug 18, 02:30**
"We turn down sales. On purpose."
Brief boutique-not-franchise . 87 min
render, \$0.99, QA clean. Posted as a
Reel to the estate page at 10:00.

 **Aug 17, 05:00**
"I wrote your laundromat resignation
letter."
Brief wd-quit-the-laundromat . 63 min
render, \$0.99, QA clean. YouTube + the
KC page.

 **Aug 17, 06:45**
"SpaceX alum builds Lego homes."
57 min render, \$0.84. **QA failed** — it
published anyway, by design. Note the
license band burned across the top.

Drift worth fixing: wd-weekly-video.sh 's own header says "one persona video a DAY ... cadence went weekly → daily 8/1", but wd-weekly-video.timer still reads OnCalendar=Mon *- * - * 05:00:00 . **The timer wins.** W/D has been shipping one video a week, not seven, since that change was written. If daily was the intent, the timer is the one-line fix; if weekly was, the script header is lying to whoever reads it next.

Identical across all four lanes

- The renderer — one file, make-product-short.mjs
- Her face — one PNG, character-ref.png
- Her voice — Jessica, cloned locally by IndexTTS-2
- Length, scene count, wardrobe rotation and camera library — all from CREATIVE-SETTINGS.env
- The #ad burn, the burned captions, the QA pass
- The Telegram ping, and the toley.io/fit link

Different per lane

- The **clock** — one timer each, deliberately staggered so only one heavy render runs at a time
- The **subject** — a database row, a rotating brief file, or a freshly built market brief
- The **link**, and whether it carries an affiliate tag at all
- The **fanout** — which channels, which accounts, which page ID
- The **AUTOPOST flag** (housing has none — it posts in-run)
- Real-estate lanes only: the burned brokerage license band

05 The lane that is *not* her — and why it matters

"New KC Homes Today" is the other real-estate video job, and people confuse the two constantly because they land on the same YouTube channel.

What listings-video actually is

A completely separate program — `housing-hub/run_listings.py` — that fires **Mon 10:30**, walks 33 KC-metro cities, collects each city's new Redfin listings through one warm browser session, verifies them in a single batched Matrix MLS pass, and builds a **photo slideshow** per city with a **Jared voice clone** narrating. No persona, no Gemini, no Wan, no Modal. Cost is essentially zero.

Cities with fewer than 2 usable listings are cleanly skipped. On the Aug 17 run that meant **6 of 33 cities shipped** — Prairie Village, Raymore, Raytown, Lone Jack, Platte City, Harrisonville — and 27 were skipped for thin inventory. One Facebook roundup post covers the day.

A separate **drain** timer at Tue 07:00 retries anything YouTube deferred for quota. That exists because a 33-city day can exhaust the daily upload quota mid-run.

Why it is worth knowing the difference

- If housing videos look wrong, **which one** matters: the lady brief (06:45, Gemini/Wan/Modal) and the listings slideshow (Mon 10:30, photos + Jared's voice) share nothing but a channel.
- Listings is **deliberately excluded** from `CREATIVE-SETTINGS.env` — its length is driven by listing count, not the 30-second ad target. Changing the ad length does not touch it.
- It is the only lane whose photos come from third-party portals, which is why the compliance choice (public portals only) is recorded in its header.



HOUSING BRIEF
The 06:45 lady lane.
License band top-left, her on camera, 30s.



0:01
Same lane, opening frame.



0:28
Same lane, close.

The license band is not decoration. Missouri 20 CSR 2250-8.070 requires the broker's licensed name on every advertisement by a licensee. It is burned **top-left**, not bottom, because YouTube's own title/CTA chrome covers the bottom ~15% of a Short — a disclosure down there is not "clearly and conspicuously displayed".

It is drawn with PIL, never `ffmpeg drawtext`, because `drawtext` silently drops one trailing character per multi-byte glyph and was truncating the license number (`KS 00251854` → `KS 002518`).

If the watermark step fails, housing refuses to publish.
That is the one hard stop in the entire system — publishing nothing beats publishing an unlicensed real-estate ad.

The daily clock, all lanes together

Time (CT)	Unit	What runs	Ends up
01:00 daily	<code>estate-video.timer</code>	1 estate video, ~87 min	queued (finished < 09:00)
02:30 daily	<code>growth-shorts.timer</code>	2 product videos, ~60–110 min each	queued
05:00 Mon	<code>wd-weekly-video.timer</code>	1 W/D video, ~63 min	queued
06:45 daily	<code>housing-hub.timer</code>	market pulse → 1 housing video → posts immediately	posted in-run
07:00 Tue	<code>listings-drain.timer</code>	retry yesterday's quota-deferred YouTube uploads	posted
10:00 daily	<code>post-pass.timer</code>	claims <code>post-queue.jsonl</code> atomically, fans every queued video out per lane	posted + logged to <code>/hq</code>
10:30 Mon	<code>listings-video.timer</code>	33 cities → per-city slideshows → YouTube + 1 FB roundup	posted

Renders were moved to the night on Aug 9 after daytime OOM lockups killed two videos mid-flight — the DGX's 128GB unified memory cannot host a Wan render and vLLM and the workers at the same time. Posting was decoupled at the same time so the overnight finish does not dump a video into a dead audience window at 4am.

06 Every call that leaves the building

Per video render. Eight outside services, plus the posting fanout. This is the complete list — if it is not here, it does not leave the DGX.

During the render

Host	What for	Stage	Cost	Credential	Notes
Neon Postgres via <code>psql</code>	Read the shop catalog — title, ASIN, images, price, haul flag	1	FREE	<code>DATABASE_URL</code> in <code>tolley-site/.env.local</code>	Treasures lane only.
amazon.com via CDP Chrome :9223	Official product gallery for the ASIN	2	FREE	Warm browser profile <code>~/research-chrome-profile</code>	Cron relaunches it via <code>systemd-run</code> if down; if it still fails, the gate starves and renders skip.
generativelanguage .googleapis.com	Vision gate — classify/reject every candidate image	2	~\$0.01	<code>~/config/autopilot.env</code>	Chain: 2.5-flash → 2.0-flash → flash-latest. All fail = heuristic pass.
Modal — Kimi K3 managed endpoint	Write the script + 7-scene storyboard (a vision model — it sees the photos)	3	\$0.20	<code>MODAL_PROXY_KEY</code> / <code>SECRET</code>	Falls back to Qwen3.6 on :8356 (free, text-only). Cron probes it with a 1-token call first.
generativelanguage .googleapis.com	Paint her into each scene — Gemini 3.1 Flash Image ("Nano Banana 2")	4	~\$0.07	<code>~/config/autopilot.env</code>	7 frames. Retries without refs on a safety refusal.
Modal — <code>lady-wan22</code> L40S GPUs	Wan2.2-S2V talking clips, one per face scene	6	\$0.12 /clip	<code>~/modal.toml</code>	The main spend. One container per clip (<code>AI_CLIP_FANOUT=1</code>) — 7 in parallel, ~8 min not ~48.
generativelanguage .googleapis.com	QA — look at 4 frames of the finished file, vote on 7 failure modes	8	~\$0.01	<code>~/config/autopilot.env</code>	Informational. Gates auto-posting, never the render.
api.telegram.org	Tell you what happened	all	FREE	<code>~/xero-ledger/.env</code>	Same bot/chat for every lane.

Wired but currently OFF

Host	Was for	Switched off	Why, and what replaced it
api.elevenlabs.io	Hosted voiceover	Aug 4	Account past-due with the character quota spent — every render paid a failed round-trip before falling back anyway. Replaced by local <code>IndexTTS-2</code> ; re-enable with <code>SHORTS_USE_ELEVENLABS=1</code> .
queue.fal.run Kling lip-sync	Mouth sync per clip	Aug 7	Your A/B call: it re-rendered every face scene at its own quality, ate ~40% of the upscale gains and inflated her mouth 24–32%, with no knobs. Superseded by S2V syncing at generation time.

At post time (10:00, or in-run for housing)

Destination	How	Which lanes
YouTube Data API	<code>content-autopilot/publishers/youtube_publisher.py</code> , OAuth refresh token per channel	Treasures + W/D → @ruthann · Housing + Listings → @yourkchomes
graph.facebook.com	Page video / Reel, page access token per page	All four, each to its own page
graph.facebook.com (IG)	Reels container + publish	Treasures
Pinterest	Local Selenium driver — :9107 Your KC Homes, :9108 Ruthann's Treasure Haul	Treasures, Housing
TikTok	Local Selenium driver on :9106 — takes a local file path, stages to the NAS outbox itself	Treasures
graph.threads.net · api.x.com · Bluesky	Direct API	Treasures
www.tolley.io/api/hq/post-log	POST with <code>x-sync-secret</code> — every leg's status, URL and error	All. This is what fills the /hq Posts tab.

07 How every link is built

Nothing in a caption is typed by hand. Every URL is assembled at metadata time from the product row, the platform, and today's wardrobe pick.

1 • The affiliate link — Treasures only

```
https://www.amazon.com/dp/B0DJ73NCSJ?tag=tolley-yt-20
```

the ASIN, straight off the `Product.amazonAsin` column · a **per-platform subtag**, so /hq can tell you which platform earned the click.

Master tag `tolley-shop-20`. Subtags: `yt / pin / fb / ig / tt / x`, each `tolley-<plat>-20`. These must be **registered in Associates Central** before that platform's leg goes live, or the commission silently drops on the floor.

2 • The storefront shelf link

```
https://www.amazon.com/shop/digitaljared/list/2XYPG23E8JCC?tag=tolley-yt-20
```

The ASIN is looked up against the committed shelf sheet to find which of the 10 Idea Lists it lives on — "Kitchen & Coffee" here — and the caption links **that shelf** instead of the bare storefront. A product not on any shelf falls back to `amazon.com/shop/digitaljared`.

3 • The shoppable-fit link — every lane

```
https://tolley.io/fit/knit-tank-midi-skirt-ivory-white-tennis-shoes-pendant-stacked-rings~pearl-hair-clips?utm_source=yt&utm_medium=social&utm_campaign=fit
```

The code is **her actual outfit for this video**, tilde-joined from the wardrobe pick: silhouette ~ colour ~ footwear ~ jewelry ~ hair. It is not stored anywhere — the page resolves the code against `persona-fit-catalog.json` at request time, which is why old videos keep working.

She also says it out loud in the last line of the script — `PERSONA_FIT_CTA=1` appends the spoken tail. The page itself is on the next spread.

4 • The brief-lane link

```
https://tolley.io/estate?utm_source=fb&utm_medium=social&utm_campaign=estate-video
```

Base URL and campaign both come out of the brief JSON, so adding a new brief lane is a file, not a code change. No affiliate tag, no `#ad` — it is your own service.

5 • The own-shop link

```
https://tolley.io/shop/cmprso4i6000mjv04ws64sgg2?utm_source=bluesky&utm_medium=social&utm_campaign=shorts
```

Used on Bluesky and Threads. Amazon Associates only permits affiliate links on FB / IG / X / YouTube / TikTok / Twitch — so those two legs point at your own page, which carries the affiliate link and the disclosure.

Compliant, and the UTM still shows in /hq referrer stats.

Why there are two link styles at all

Amazon Associates only permits affiliate links on FB, IG, X, YouTube, TikTok and Twitch. Bluesky and Threads are not on that list — so those legs link to `tolley.io/shop`, which carries the affiliate link and the disclosure itself. Fully compliant, and the UTM still shows up in /hq referrer stats.

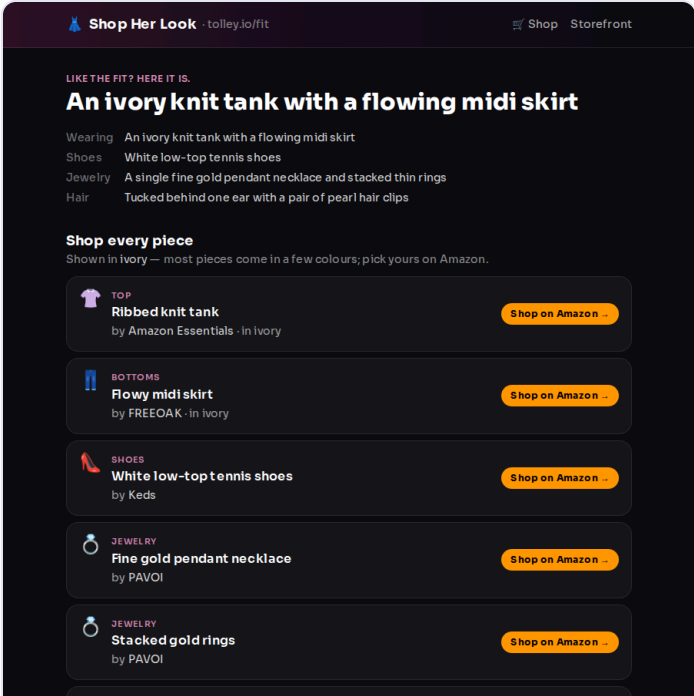
Instagram is a third case: it permits the link but the platform gives you nowhere clickable to put it, so the caption says "link in bio" and the affiliate URL rides in the metadata as `referenceLink` for the manual bio/Creator-Center pass.

What is *not* automated

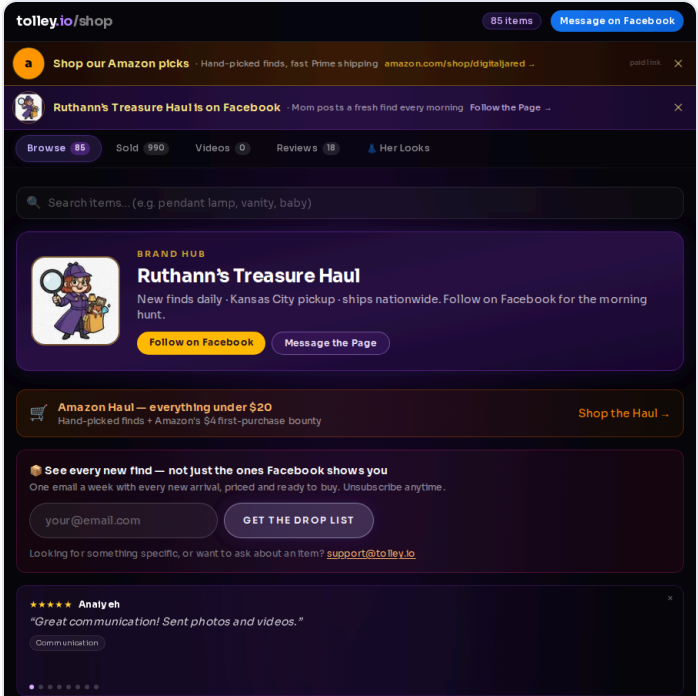
- **TikTok product anchors.** The clickable path is a Creator Center "Link Products" pass — the caption can only point at the bio.
- **The IG bio link** itself.
- **Registering a new subtag** in Associates Central before that platform's leg goes live. An unregistered subtag silently earns nothing.

07b Where those links land

Both live right now. Left: the exact fit code generated by the Aug 18 product video. Right: the shop page the non-Amazon platforms point at.



toley.io/fit/knit-tank-midi-skirt-ivory~... — every piece resolves to a real Amazon product with the affiliate tag attached, plus the disclosure block. The code is resolved at request time, so old videos keep working forever.



toley.io/shop — where the Bluesky, Threads and "link in bio" legs land. It carries the affiliate links and the disclosure, which is what keeps those platforms compliant with Associates' platform whitelist.

The full round trip of one click

- 1. Viewer sees the video on **YouTube**, hears "the link for that is below too".
- 2. Description carries four links: the product with toley-yt-20 , the shelf, toley.io/shop , and the fit code with utm_source=yt .
- 3. Click the fit link → **toley.io** logs the visit with its UTM → /hq Site tab.
- 4. Click "Shop on Amazon" → Amazon with the tag → commission lands under that subtag in Associates.
- 5. The posting run already wrote the video's URL and status into the **/hq Posts tab** via api/hq/post-log ; the view counter picks it up from there.

That is the whole attribution chain: subtag = platform, UTM = page, post-log = video. Nothing else is instrumented.

Where the disclosure goes, per platform

Platform	Link style	Disclosure
YouTube	Full affiliate + shelf + shop + fit	Line 2 of the description
Facebook	Affiliate + fit	Caption + burned #ad
Instagram	No link — "link in bio"	In caption
Pinterest	Affiliate as the pin destination	Truncation protects the disclosure, never the hook — an Amazon link with no disclosure is a ToS strike
TikTok	Text pointer only (captions can't carry links)	In caption
X	Affiliate, toley-x-20	#ad (amazon affiliate)
Bluesky / Threads	toley.io/shop only	Plain #ad

Two disclosures, always. One burned into the pixels for the whole 30 seconds, one in the caption text. Neither is optional and neither is conditional — the burn happens in ffmpeg before the file is written, so a video physically cannot exist without it.

08 What it costs, and where

Figures are the renderer's own ledger, written into every video's JSON and pushed to the /hq Posts tab by `push-video-costs.mjs` after each run.

\$0.84

Wan clips
7 × \$0.12 on Modal L40S

\$0.20

Kimi K3
script + storyboard

~\$0.09

Gemini
gate + 7 keyframes + QA

\$0.00

voice, lip-sync,
assembly, captions

≈ \$1.13 a video, ≈ \$170 a month

Lane	Per week	Per month
Treasures (2/day)	14	~61 · \$69
Estate (1/day)	7	~30 · \$34
Housing (1/day)	7	~30 · \$32
W/D (1/week)	1	~4 · \$5
Listings (Mon, ~6 cities)	~6	~26 · ≈\$0
Total	~35	~151 videos · ~\$140/mo

Compare: TubeGen quotes \$1.94–2.47 *per minute*, Vidnoz \$26.99/mo for 15 minutes total. At 151 × 30s you produce ~75 minutes a month for ~\$140 — and you own the face, the voice and the wardrobe.

What the Aug 8 cost ruling actually did

- **45s → 30s**: 10 scenes → 7. Fewer paid clips *and* fewer paid keyframes. ~30% off.
- **720p → 480×720**: ~\$0.17 → ~\$0.07 per i2v clip.
- **12 → 8 sampler steps**: another ~33% off every clip and every retry.
- **Face gates off**: they were spending up to 12 extra paid generations per video.
- **Lip-sync → local**: was ~\$0.90/video on Kling, now \$0.
- **Enhance + face-restore off**: no cash, but it gave back ~20 min of DGX GPU per video.

Where the *time* goes — and it is not the money

Stage	Wall clock	Cash
Pick + images + vision gate	~5 min	~\$0.01
Kimi script + storyboard	~2 min	\$0.20
7 Gemini keyframes	~4 min	~\$0.07
Voice + fit	~3 min	\$0
7 Wan clips	~45 min	\$0.84
Assembly + captions + mux	~4 min	\$0
QA	~1 min	~\$0.01
Measured, Aug 18	66 min	\$1.04

Observed renders this week: 57, 63, 66, 87 min. The spread is almost entirely Modal queue time on the clip fan-out.

The lever nobody uses. `WAN_BACKEND=local` is already set — image-to-video clips prefer the DGX's own GPU on `:8190` for **\$0** and only fall out to Modal per-clip on failure. But `SHORTS_ANIMATION=auto` sends every *face-visible* scene to **S2V, which has no local path** — and in the premium format she is in every scene. So in practice all 7 clips bill to Modal at \$0.12 and the free local path never runs.

That is your single biggest cost decision: \$0.84/video buys generation-time mouth sync. `SHORTS_ANIMATION=off` reverts to local i2v + local LatentSync — roughly **\$0.00** in clips — at the cost of slower renders and a separate sync pass. Worth an A/B on one night's output.

09 The knobs — where to change what

One file drives creative decisions across all five pipelines: `~/business-os/CREATIVE-SETTINGS.env` . A real environment variable of the same name always wins, so any wrapper can still override per-run.

Change these and every lane changes tonight

Setting	Now	What it actually controls	If you change it
AD_VIDEO_TARGET_SECONDS	30	Video length — and therefore scene count, and therefore cost	45 → 10 scenes, ~+43% cash. 88 → 18 scenes. Hard-capped at 89s so FB/IG still auto-route to Reels.
SHORTS_ANIMATION	auto	Motion engine per scene. auto = S2V for face shots (Modal, paid), i2v for the rest	off = pure local i2v + LatentSync, near-\$0 clips, slower, separate sync pass
WAN_WIDTH × HEIGHT	480×720	Wan render resolution before the 1080×1920 upscale	720×1280 ≈ 2.7× pixels ≈ 2–2.5× clip cost. This is most of why clips read soft.
WAN_STEPS	8	Sampler steps. Buys cleaner motion (fewer torn limbs), not more detail	12 was the old default — ~+50% clip cost
PERSONA_OUTFIT_ROTATION	1	New outfit every video from <code>persona-wardrobe.json</code>	0 = she reverts to the olive dress in the reference photo, forever
PERSONA_GLAM_EVERY_N	4	Every 4th video pulls from the glam list instead	0 or 1 disables glam; standard rotation keeps running
PERSONA_CAMERA_ROTATION	1	Real framing/lens/angle per scene + a genuine camera move on the clip	0 reverts to constant three-quarter framing and no movement
PERSONA_FIT_CTA	1	The spoken "like the fit?" tail + the <code>tolley.io/fit</code> caption link	0 removes both. Note it costs you ~2s of the 30.
SHORTS_FACE_GATES	off	Re-roll a scene when her eyes/mouth/extra-people check fails	audit = measure only. on = up to <code>SHORTS_RETRY_TOTAL_CAP</code> extra <i>paid</i> generations per video
SHORTS_ENHANCE	0	RIFE 4.9 interpolation + RealESRGAN upscale, locally, for \$0	1 = measurably sharper (duplicate frame transitions 21% → 0%) at ~+13 min GPU per video
SHORTS_FACE_RESTORE	0	GFPGAN pass <i>after</i> lip-sync — the ordering fix for "eyes are a bit blurry"	1 at weight 0.7 = +23% eye detail. Do not exceed 0.8 or she goes porcelain.
VIDEOS_PER_DAY	2	Treasures renders per night (env, set in the cron)	3 ≈ +\$34/mo. Each one adds ~1 hour of overnight GPU.

Content files — edit these, not the code

<code>persona-wardrobe.json</code>	every outfit she can wear · <code>~/business-os/</code>
<code>persona-cinema.json</code>	every camera move and framing · <code>~/business-os/</code>
<code>character-ref.png</code>	her face. One file, all four lanes.
<code>estate/briefs/*.json</code>	30 estate scripts, day-of-year rotation
<code>wd-content/briefs/*.json</code>	30 W/D scripts, same rotation
<code>persona-fit-catalog.json</code>	what each fit code resolves to on <code>tolley.io/fit</code>
<code>gate-failed.json</code>	ASINs benched for 14 days after a failed image gate
<code>posted.json</code>	what has already been made, so nothing repeats

Adding a brief is genuinely just dropping a JSON file in the folder — slug, title, pitch, offer, link, campaign, hashtags. The rotation picks it up the next day it comes round.

The three flags that decide "draft or live"

`growth-engine/shorts/AUTOPOST` **present** ✓

`growth-engine/estate/AUTOPOST` **present** ✓

`growth-engine/wd-content/AUTOPOST` **present** ✓

All three lanes are **live** right now. Delete the file and that lane renders to `review/` and Telegrams you a draft instead. touch it to go live again. Nothing else changes.

The one you cannot flip from a file: housing posts *in-run* at 06:45, not through the 10:00 queue, so it has no AUTOPOST flag. It is gated by `HOUSING_LINKEDIN` / `HOUSING_PINTEREST` per leg only.

10 Where it breaks, and what to type

The failure modes that have actually happened

Symptom	Cause & the guard that now exists
Lane renders 0 of 2, two nights running	The picker kept retrying the same 8 products that failed the image gate while 70+ untried ones waited. Fixed Aug 16 with the 14-day bench.
Every video exactly 45.0s with dead air at the end	A length floor collided with the target when ads were cut to 45s — a 34s read padded to 45s while she kept mouthing. The floor is gone; length follows the narration.
A ghost of her face slides across a cut	A dissolve between two face shots superimposes two versions of her. Two adjacent face shots are now always a hard cut.
Smeared, torn, duplicated arms	Fast limb motion. The storyboard prompt now forbids waving/throwing/sweeping and anything passing near the lens.
A random man appears in her scene	Wan hallucinating a second person. Fixed in the negative prompt — "second person, another person, disembodied hands".
Psychedelic mush at the end of a clip	Wan drifting over the last ~10 frames. The FLF2V end-frame anchor (AI_CLIP_HOLD) holds it.
Renders die mid-flight in the afternoon	OOM — the DGX cannot host a Wan render, vLLM and the workers in 128GB at once. Renders moved to 01:00–06:45 on Aug 9.
License number truncated to KS 002518	ffmpeg drawtext drops a trailing char per multi-byte glyph. The band is drawn with PIL now.
Nothing renders at all	Both script writers down. The cron probes Kimi, tries docker start vllm_qwen36 , and only then gives up with a Telegram.

The structural risk to watch. Because every stage degrades instead of failing, a video can ship with 3 of 7 scenes as Ken Burns stills and a QA "pass". SHORTS_DEGRADED_NOTE_AT=3 Telegrams you when that happens — worth confirming you actually receive those. It is the only signal separating "the lane is working" from "the lane is running".

Copy-paste

Render one video right now, by hand

```
node ~/growth-engine/shorts/make-product-short.mjs --auto \
  --outdir ~/growth-engine/shorts/review
```

Render a specific product

```
node ~/growth-engine/shorts/make-product-short.mjs \
  --product-id <id> --template punch-cut
```

Render a specific brief

```
~/growth-engine/estate/estate-video.sh \
  ~/growth-engine/estate/briefs/<slug>.json
```

Post something that is sitting in review/

```
python3 ~/growth-engine/shorts/post-short.py \
  <video.mp4> <meta.json> --tt-account ruthann
```

Retry one failed leg only

```
python3 ~/growth-engine/shorts/post-short.py \
  <video.mp4> <meta.json> --only ig
```

Go draft / go live

```
rm ~/growth-engine/shorts/AUTOPOST # draft
touch ~/growth-engine/shorts/AUTOPOST # live
```

Watch tonight's run

```
tail -f ~/growth-engine/shorts/logs/$(date +%F).log
journalctl --user -u growth-short -f
```

What is queued for 10:00

```
cat ~/growth-engine/post-queue.jsonl | jq .
```

Force the post-pass early

```
systemctl --user start post-pass.service
```

See every timer

```
systemctl --user list-timers | grep -E \
  'estate|shorts|wd-|housing|listings|post-pass'
```

Three things worth doing next

- 1. **Settle the W/D cadence.** The script says daily, the timer says Monday. One of them is wrong and has been for two weeks.
- 2. **A/B SHORTS_ANIMATION=off for one night.** It is the difference between ~\$1.04 and ~\$0.20 a video. If the local i2v + LatentSync output is acceptable, that is ~\$100/mo back.
- 3. **Turn SHORTS_ENHANCE back on.** It costs GPU time, not money, and the measured gain (0% duplicate frame transitions vs 21%) is the cheapest quality you have available.

Sources. Everything here was read off the DGX on Aug 18, 2026: growth-engine/shorts/make-product-short.mjs (2,550 lines) · daily-short-cron.sh · estate/estate-video.sh · wd-content/wd-weekly-video.sh · post-pass.sh · lib/persona-wardrobe.mjs · lib/postlog.sh · shorts/{scene_frames,qa_check,classify_images,animate-clips,post-short}.py · housing-hub/{run_daily,run_listings,housing_short}.py · business-os/CREATIVE-SETTINGS.env · systemctl --user list-timers · the render JSON of the Aug 17–18 output of every lane · and a surviving working directory (/tmp/short-EdIyZh) for the stage-by-stage frames.